

M003 Week 2 — English Worksheet (Beginner)

Topic: Lines & Rectangles on the Wall (x, y) · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

This week you use a **loop** to draw a whole **line** of blocks on the wall instead of placing them one at a time. Grow **x** to go across. Grow **y** to go up.

1 · Predict 🧠

Read the steps. Before you imagine it happening, circle or write your answer.

```
set x to 0
repeat 4 times:
  place stone block at (x, 0)
  add 1 to x
```

x grows and y stays 0. Does this draw a line going across, or a line going up? Circle one: across · up

Why?


```
set y to 0
repeat 4 times:
  place stone block at (0, y)
  add 1 to y
```

This time y grows and x stays 0. Across or up? Circle one: across · up

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — The line should go **across** the bottom of the wall.

```
# clean
set x to 0
repeat 4 times:
  place block at (x, 0)
  add 1 to x
```

```
# buggy
set x to 0
repeat 4 times:
  place block at (0, x)
  add 1 to x
```

What is wrong? Which way does the buggy line go?

Pair B — The line should be **4 blocks long**.

```
# clean
set x to 0
repeat 4 times:
  place block at (x, 0)
  add 1 to x
```

```
# buggy
set x to 0
repeat 2 times:
  place block at (x, 0)
  add 1 to x
```

What is wrong? How long is the buggy line?

3 · Fill the Gap

You want a line going **straight up** the left edge of the wall. x must stay **0** while y grows. One number is missing. Fill it in using the word bank.

```
set y to 0
repeat 4 times:
  place block at (____, y)
  add 1 to y
```

Word bank: 0 · y · 4

Write the missing number:

4 · Tell Me What You Built 📷

Now switch to your homework world. Use a loop to draw **one line going across** and another loop for **one line going up**, so they meet at the corner (0, 0) and make an **L** shape. When you finish, come back here.

Send a photo or video of your L, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

My across line grew the number ...

My up line grew the number ...

The two lines meet at ...

5 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) while you show your L on the wall. Talk like you are teaching a friend. Try to use these words: **loop**, **line**, **across**, **up**.

1. Run the loop that draws the across line.
2. Run the loop that draws the up line.
3. Say which number (x or y) grew in each loop.

Write what you will say in your video. You can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.